

Skills Summary

One-Variable Inequalities: The Sign-Flip Rule

Use this reference while completing your worksheet or when studying.

Solving One-Step Inequalities

Rule: solve an inequality with the same moves as an equation, with **one exception** — multiplying or dividing both sides by a **negative** number **reverses** the inequality symbol ($< \leftrightarrow >$ and $\leq \leftrightarrow \geq$). Adding or subtracting never reverses it.

Example: Solve $-6x < 24$

Step 1. The variable is multiplied by -6 , so divide both sides by -6 — and because -6 is negative, reverse the symbol.

Step 2. $x > \frac{24}{-6}$, so $x > -4$.
 $\Rightarrow x > -4$

Try it: Solve $-8x \geq 32$

check: $-4 \leq x$

Solving Two-Step Inequalities

Order of moves: undo the adding or subtracting first, then undo the multiplying or dividing. Only the *last* move can trigger the sign-flip rule, and only when the number you divide by is negative.

Example: Solve $-4x + 3 > 15$

Step 1. Undo the $+3$ first (subtracting never flips the symbol), then divide by the negative coefficient and reverse.

Step 2. $-4x > 15 - 3$, so $-4x > 12$; dividing by -4 gives $x < -3$.
 $\Rightarrow x < -3$

Try it: Solve $-5x - 2 \leq 13$

check: $x \geq -3$

Catching a Missing Sign Flip

Checklist for grading your own work:

- Did any step multiply or divide by a negative? Then the symbol must have reversed in that step.
- Test one easy number from your answer in the *original* inequality — if it makes the inequality false, the symbol is pointing the wrong way.

Example: Jo says $-2x < 10$ means $x < -5$. Check Jo's claim.

Step 1. Jo divided by -2 , a negative, so the symbol should have reversed — test a number from Jo's answer to confirm the claim fails.

Step 2. Try $x = -6$ from Jo's set: $-2(-6) = 12$, and $12 < 10$ is false. Solving correctly: divide $-2x < 10$ by -2 and reverse.

$$\Rightarrow x > -5$$

Try it: Ali says $-3x \geq 21$ means $x \geq -7$. Fix it.

check: $-7 \geq -7$